

INTERNSHIP REPORT

On

AI Resume Screening & Job Match System

Submitted To

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of
B.Tech CSE (Data Science and AI)

By

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CANDIDATE'S DECLARATION

I, ANUSHKA, do hereby declare that the internship report titled "AI Resume Screening & Job Match System" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the internship program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the Ybi Foundation team for the opportunity to work as a Data Science and Machine Learning Intern and for their guidance throughout the internship period.

I express my gratitude to my parents and faculty at IILM University for their constant support and encouragement.

Signature

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INTERNSHIP COMPLETION CERTIFICATE

The Internship Completion Certificate issued by Ybi Foundation (Credential ID: QZEKFSSBI5L7M), confirming completion of the internship in Data Science and Machine Learning for a duration of 2 Weeks, concluding on 26 June 2026, is attached below.

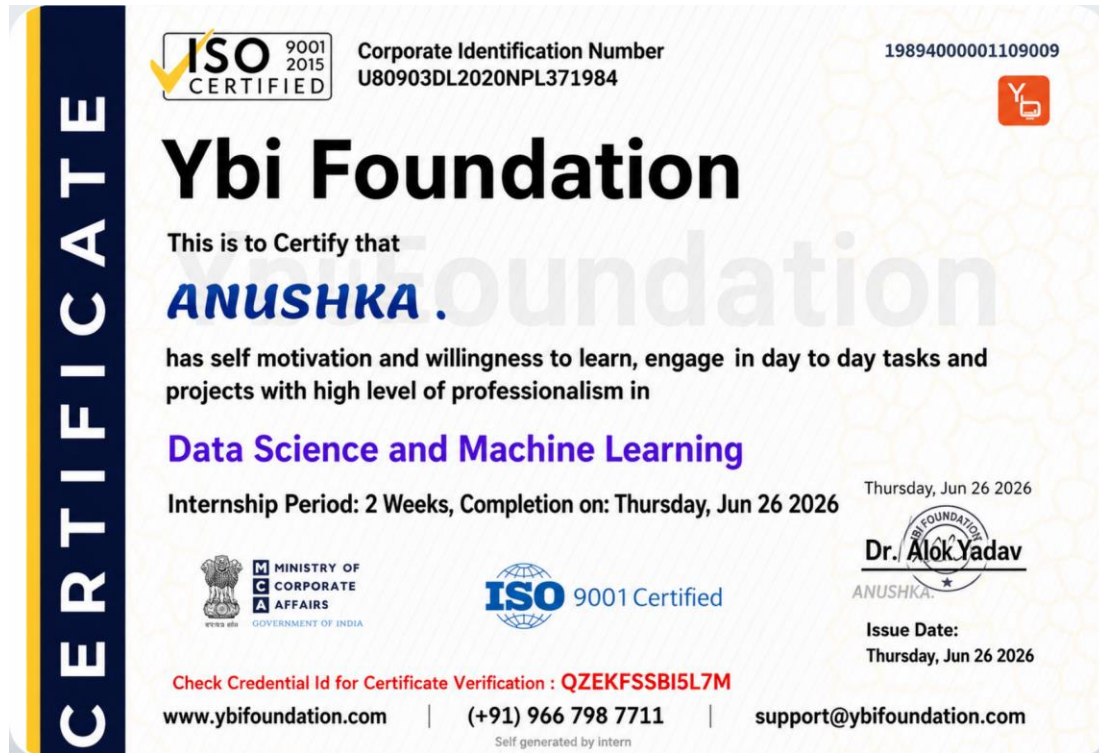


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4. PROJECT DESCRIPTION

4.1 Introduction

This report presents the work carried out during my internship as a Data Science and Machine Learning Intern at Ybi Foundation, undertaken as part of the internship requirement for the degree of Bachelor of Technology in Computer Science and Engineering (Data Science and AI) at IILM University, Greater Noida. The internship, spanning a duration of 2 Weeks and concluding on 26 June 2026, focused on applying data science and natural language processing techniques to build a practical, real-world solution.

During the internship, I designed and developed the "AI Resume Screening & Job Match System", an application that automates the comparison of a candidate's resume against a given job description. The system extracts text from an uploaded resume, processes both the resume and the job description using natural language processing techniques, and computes a job match percentage using similarity measures. The tool reflects the practical use of NLP and machine learning to solve a real problem faced by both job seekers and recruiters — quickly and objectively judging how well a resume fits a role.

4.2 Organization Profile

Ybi Foundation is the organization that hosted the internship, offering ISO 9001:2015 certified internship programs to students seeking practical, industry-facing experience in Data Science, Machine Learning, and related technology domains. The internship completion certificate (Credential ID: QZEKFSSBI5L7M) confirms successful completion of the internship in Data Science and Machine Learning. Further organizational details, such as specific team structure and mentors, can be added here from information provided directly by Ybi Foundation.

4.3 Problem Statement

Recruiters receive a large number of resumes for every open position, making manual screening slow, inconsistent, and prone to human bias. At the same time, job seekers often struggle to know how well their resume matches a particular job description, or which skills they are missing for a role they are targeting. There was a need for a system that could automatically read a resume, compare it against a job description, and objectively report a match percentage along with matching skills, missing skills, and suggestions for improvement — without requiring manual comparison of the two documents.

4.4 Project Objectives

- Allow a user to upload a PDF resume and paste or enter a job description for comparison.
- Extract clean text content from the uploaded resume using PDF parsing techniques.
- Process both the resume and job description using NLP techniques to represent them in a comparable form.
- Calculate a Job Match percentage between the resume and job description using a similarity measure.
- Identify and display matching skills and missing skills between the resume and job description.

- Suggest a relevant job role based on the resume content.
- Visualize the skill comparison using charts, and provide resume improvement suggestions.

4.5 Scope of the Project

The scope of the project covers reading a PDF resume, extracting its text, and comparing it against a user-provided job description using NLP-based similarity techniques. The system computes a match percentage, lists matching and missing skills, suggests a probable job role, and presents the comparison visually through charts, along with resume improvement suggestions. The project is limited to text-based resumes in PDF format and does not cover applicant tracking system (ATS) integration, resume writing, or automated job applications; those are identified as areas for future enhancement.

4.6 Technologies and Tools Used

- Python – core programming language for the project.
- Pandas – for handling and organizing extracted data.
- Scikit-learn – for implementing NLP and similarity-based ML techniques.
- NLP / TF-IDF (Term Frequency–Inverse Document Frequency) – for converting resume and job description text into comparable vector representations.
- Cosine Similarity – for calculating the Job Match percentage between the resume and job description vectors.
- PyPDF2 / PyMuPDF – for extracting text content from uploaded PDF resumes.
- Streamlit – for building and deploying the interactive web application.
- Matplotlib / Plotly – for generating the skill comparison charts.

4.7 System Architecture

Figure 4.1 – AI Resume Screening & Job Match System: PDF Resume Upload → Text Extraction (PyPDF2/PyMuPDF) → NLP Preprocessing of Resume and Job Description → TF-IDF Vectorization → Cosine Similarity Calculation → Job Match % → Skill Comparison (Matching/Missing Skills) → Suggested Job Role and Resume Improvement Suggestions → Results displayed on the Streamlit interface with charts.

Component	Role
Resume Upload (Streamlit)	Accepts the PDF resume uploaded by the user
Text Extraction (PyPDF2/PyMuPDF)	Extracts raw text content from the uploaded PDF resume
NLP Preprocessing	Cleans and prepares resume and job description text for comparison
TF-IDF + Cosine Similarity (Scikit-learn)	Converts text to vectors and computes the Job Match percentage
Skill Comparator	Identifies matching and missing skills between resume and job description
Visualization (Matplotlib/Plotly)	Generates the skill comparison chart
Recommendation Engine	Suggests a job role and resume improvement tips

4.8 Methodology

The project was approached step by step. First, a Streamlit-based interface was built to let the user upload a PDF resume and enter or paste a job description, with an "Analyze Resume" action to trigger processing. PyPDF2/PyMuPDF was used to extract text from the uploaded resume PDF. Both the resume text and job description were then preprocessed using standard NLP techniques (tokenization, stop-word removal, and normalization) before being converted into TF-IDF vector representations using Scikit-learn.

Cosine similarity was applied between the resume and job description vectors to compute the Job Match percentage. Skill sets were extracted from both texts and compared to identify matching skills (present in both) and missing skills (present in the job description but absent from the resume). Based on the extracted skills, a suggested job role was generated, and simple resume improvement suggestions were produced by highlighting the missing skills and phrasing gaps. Finally, Matplotlib/Plotly was used to render a skill comparison chart, and all outputs — match percentage, matching skills, missing skills, suggested role, chart, and improvement suggestions — were displayed together on the Streamlit results page.

4.9 Features and Expected Outcomes

- ☐ Match Percentage – an overall score showing how well the resume matches the job description.
- ☒ Matching Skills – skills found in both the resume and the job description.
- ☒ Missing Skills – skills required by the job description but not found in the resume.
- ☐ Suggested Job Role – a probable job role based on the resume's content.
- ☐ Skill Comparison Chart – a visual comparison of matching vs. missing skills.
- ☐ Resume Improvement Suggestions – practical tips to improve the resume's match with the target role.

Overall, the project provides a working, interactive tool that reduces the manual effort of comparing a resume against a job description, gives job seekers actionable feedback to improve their resumes, and offers a foundation that could be extended toward ATS-style resume screening for recruiters, along with hands-on experience applying NLP and machine learning techniques (TF-IDF, cosine similarity) within a deployed web application.

4.10 Certificate of Completion

The Ybi Foundation Internship Completion Certificate (Credential ID: QZEKFSSBI5L7M) is attached in Section 3 of this report as proof of successful completion of the internship.

5. BIBLIOGRAPHY/REFERENCES

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